Intro to psycholinguistics

Day 4: Recognizing a stream of sounds as (one's) language

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Setting the stage

Babies go from (next to) zero language to language in a short time.

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For lexical and syntactic *production*, the typical development path goes like this:

- ✧ Cooing and babbling < 8 mo
  vocal play, vowels, consonants, syllables
- ✧ One-word stage < 18 mo
- ✧ Two-word stage < 24 mo
  e.g., “Ken water” for “Ken is drinking water”
- ✧ Early multiword stage < 30 mo
  e.g., “Want lady get chocolate” (23 mo)
- ✧ Later multiword stage
  e.g., “I’m having this little one. Me’ll have that.” (30 mo)

Source: Liberman lecture notes.

Babies go from (next to) zero language to language in a short time.

Milestone	**Nelson 1973** (18 children)	**Fenson 1993** (1,789 children)
10 words	15 months (range 13-19)	13 months (range 8-16)
50 words	20 months (range 14-24)	17 months (range 10-24)
Vocabulary at 24 months	186 words (range 28-436)	310 words (range 41-668)

vocabulary size by age

Source: Liberman lecture notes.

Babies go from (next to) zero language to language in a short time.

```
CHILD: My teacher holded the baby rabbits and we patted them.
ADULT: Did you say your teacher held the baby rabbits.
CHILD: Yes.
ADULT: What did you say she did?
CHILD: She holded the baby rabbits and we patted them.
ADULT: Did you say she held them tightly?
CHILD: No, she holded them loosely.
```

example of overregularization

Source: Liberman lecture notes.

Babies go from (next to) zero language to language in a short time.

Babies acquiring any language, even sign languages, all go through the same stages at approximately the same times.

For *perception*, they are born already knowing a lot.
Language acquisition appears to start in the womb.

- ✧ Newborns can recognize their birth mother's voice, their native language, some phonetic contrasts (e.g., [ba] vs. [da]).
- ✧ $\sim$ 6 month olds can discriminate more phonetic contrasts, even ones from non-native languages. This gets lost in time.
- ✧ By 7.5 months old, infants can detect individual words in fluent speech.
- ✧ By 9 months old, stress and other prosodic patterns ('robin' vs. 'giraffe') and rules governing sequences of sounds.

  ...

Source: Jusczyk & Aslin (1995).

How??

Learning language is hard

Regarding the *sound (structures)* of their language…

- ✧ What do babies need to learn?
- ✧ What are some problems they need to solve?

Learning language is hard

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- Which sounds belong to the language(s) that I'm born into?

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  (Words are not frequently presented in isolation to infants.)

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What else?

Today

We won't answer all of these questions, but we will see:

- What are the methods used to study what infants know about language?
- What are some of the things that they know about language, even before they are able to produce it?

Videos and activities

- Which ones of these problems persist for learning languages as a grown up?
- Finding word boundaries is hard!
  `Try it out.`
- Sign language babbling
  `https://youtu.be/ChsODznkINQ`

Topics in language acquisition

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We won't fully answer the *how* question.

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To do this, we focus on how very young children…

- ✧ recognize some acoustic input as (their own) language,
- ✧ figure out what words mean (next week).

Mehler et al. (1988)

A precursor of language acquisition in young infants*

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Abstract

Four-day-old French and 2-month-old American infants distinguish utterances in their native languages from those of another language. In contrast, neither group gave evidence of distinguishing utterances from two foreign languages. A series of control experiments confirmed that the ability to distinguish utterances from two different languages appears to depend upon some familiarity with at least one of the two languages. Finally, two experiments with low-pass filtered versions of the samples replicated the main findings of discrimination of the native language utterances. These latter results suggest that the basis for classifying utterances from the native language may be provided by prosodic cues.

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General problems

- What are the two problems that an infant faces in recognizing something they hear as language?

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- What questions does the paper raise, and what is their answer?

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- What are the two problems that an infant faces in recognizing something they hear as language?
- What questions does the paper raise, and what is their answer?
- Were there aspects of the paper that you found difficult?

Questions about methods

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- ✧ What were the populations tested?
- ✧ What were the two experimental tasks used? Why these tasks?
- ✧ What are the independent and the dependent variables used in Experiment #⟨pick one⟩?

An **independent variable** is one that is thought to have an effect on some measure, the **dependent variable**. In an experiment, changes are made to the IV by the experimenter, and whether or not these changes affect the DV is measured.

Notions in experimental methodology

Using infant behavior to probe hidden psychological mechanisms

Up to a certain age, it's not possible to study infants' linguistic output or to ask them to do things.

But, there are a few things that infants can do:
Sucking, **looking at things** and **turning their heads**.

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To study infants' linguistic knowledge, researchers

- ✧ can manipulate experimental variables,

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- ✧ can manipulate experimental variables,

  e.g., playing Mom's voice vs. someone else's, French vs. Russian sentences, a /b/ sound vs. a /d/ sound, …
- ✧ and see if these changes affect infants' behavior,

  e.g., do they suck faster, slower or the same? where do they look and for how long?

If an infant reliably responds differently to, e.g., French vs. Russian, we conclude that they know that they are different.

Testing babies' linguistic knowledge

High amplitude sucking paradigm

- Infants are fitted with a pacifier and headphones.
- The pacifier is connected to a device that measures the frequency and intensity of sucks ("non-nutritive sucking").
- Different stimuli are presented auditorily.
- Changes in sucking frequency or intensity are recorded.

- **Familiarisation phase**:
  One kind of stimulus is presented until the infant gets used to it.
- **Test phase**:
  The same or a different kind of stimulus is presented.



What it looks like: `video`

For more on sucking paradigms:

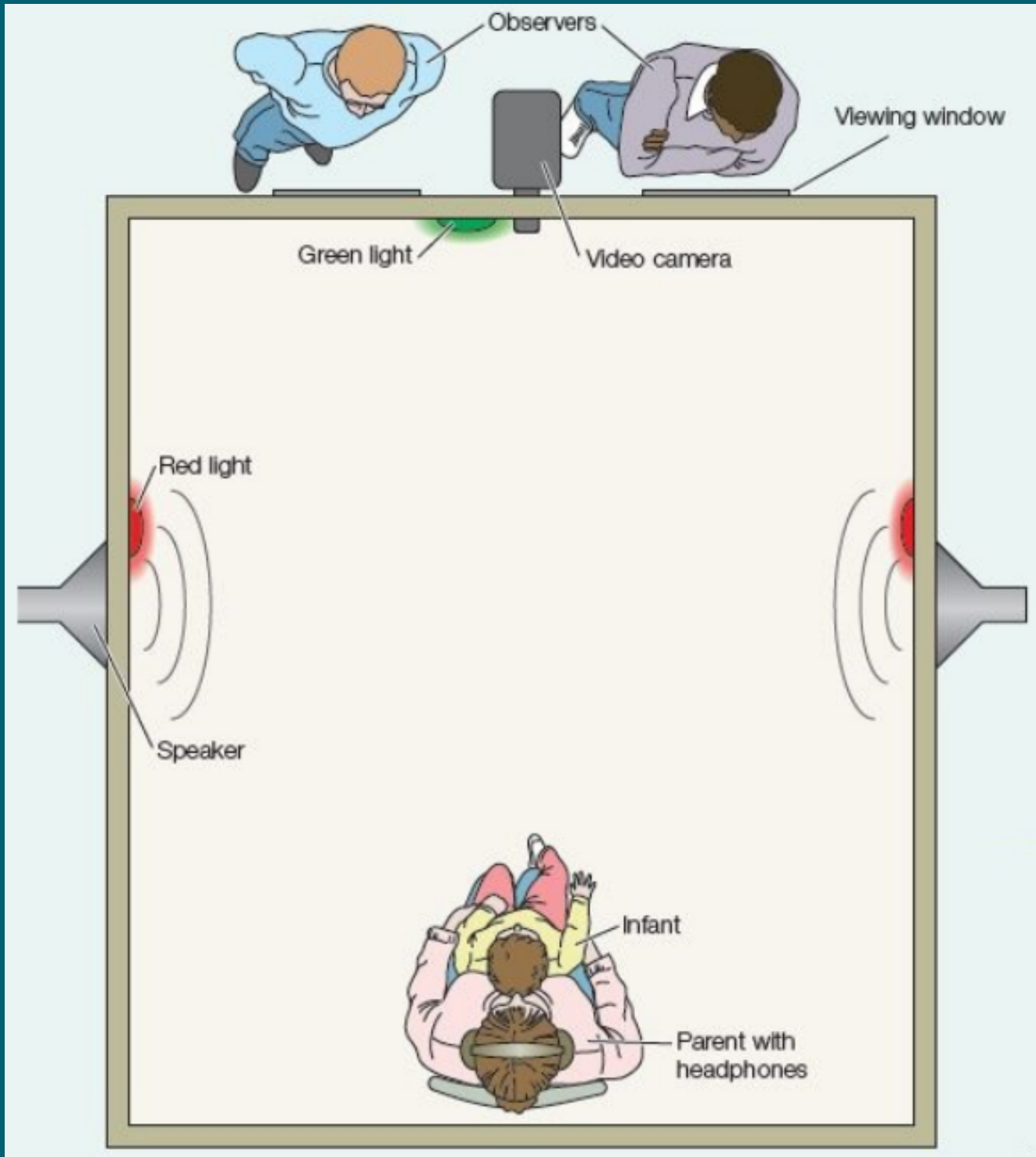
- The contingent sucking rate paradigm
  `online lecture 1 (Isabel McKay)`
- The high amplitude sucking paradigm
  `online lecture 2 (Isabel McKay)`

Testing babies' linguistic knowledge

Head turn preference paradigm

- Infants are sat on a caretakers lap.
- Sounds are presented from left or the right of the infant.
- The length of time they trun their heads in the direction of a sound is measured.

This procedure also involves a familiarization phase and a test phase.


Observers
Viewing window
Green light
Video camera
Red light
Speaker
Infant
Parent with
headphones

What it looks like: `video`

Linking hypotheses

To conclude anything from the relationship between (changes in) what infants do and what they know about language, we need some **linking hypotheses**.

> *[W]henever two stimuli cause physically indistinguishable signals to be sent from the sense organs to the brain, the sensations produced by those stimuli, as reported by the subject in words, symbols or actions, must also be indistinguishable.*
>
> *Brindley (1960), via Teller (1984)*

$\approx$ If you react differently to two stimuli (e.g., sounds of French vs. Russian), it's that your mind/brain treats them differently.

Linking hypotheses

Usually, infants react to things differently along two dimensions:

- Is a stimulus **new or old**?

  If new, I'm surprised/pay more attention to it.
  If old, I get bored/pay less attention to it.

- Is a stimulus **pleasant or not**?

  If pleasant, I like it/pay more attention to it.

More attention $\Rightarrow$ more sucking, head turns, longer gazes

These two factors might interact: Familiar stimuli might be boring (less attention) or pleasant (more attention).

Back to Mehler et al.'s experiments

Experiments 1 & 4

4 day old and 2 month old infants are able to distinguish their language from a foreign one.

- 4 day olds: French vs. Russian
- 2 month olds: English vs. Italian

4 day olds also show a preference for listening to their language that the 2 month olds don't.

A closer look at Experiment 1 (Mehler et al. 1988)

Participants: 40 4-day-old French infants.

Materials: 15 French sentences and 15 Russian sentences recorded by the same bilingual speaker.

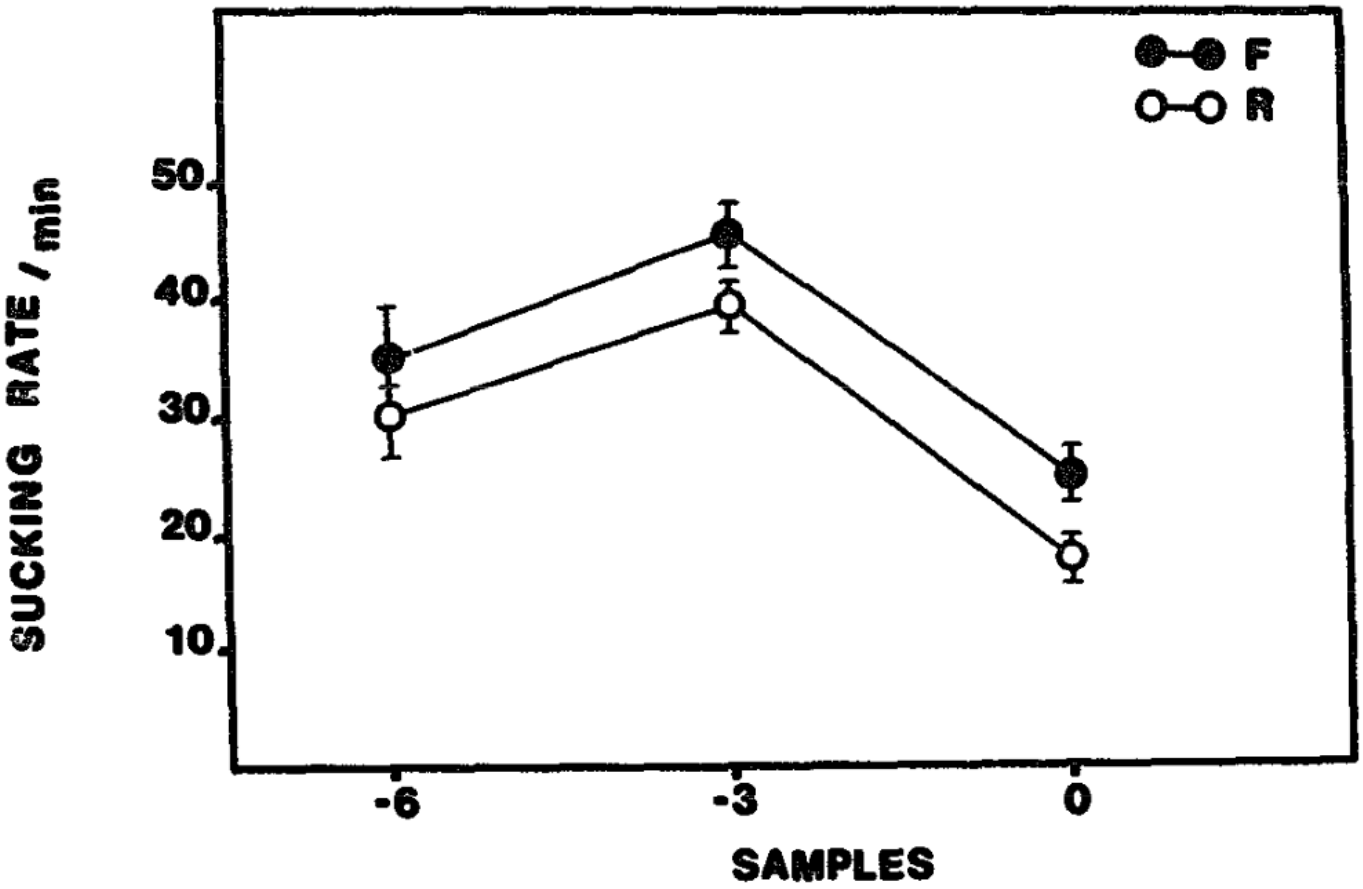
Conditions (10 infants per condition):

Habituation phase	Test phase
French	French
Russian	Russian
French	Russian
Russian	French

Procedure: A version of the High Amplitude Sucking Paradigm.

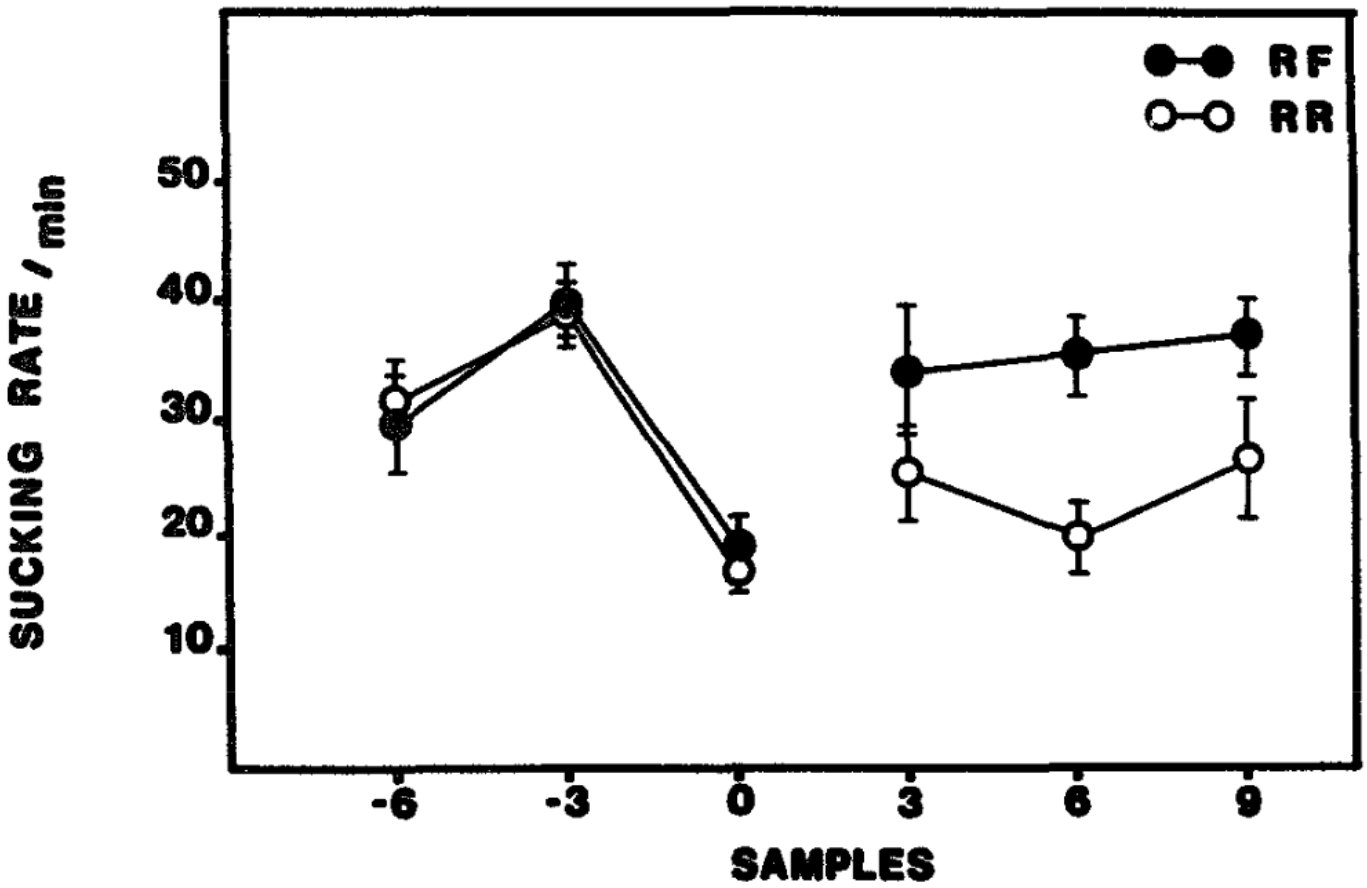
A closer look at Experiment 1 (Mehler et al. 1988)

Figure 1. *Displays the change in sucking rate for the last three blocks of three consecutive samples during the first phase of Experiment 1 for the infants who heard French (F) and the infants who heard Russian (R). The bars above and below each point indicate the standard error of the mean.*

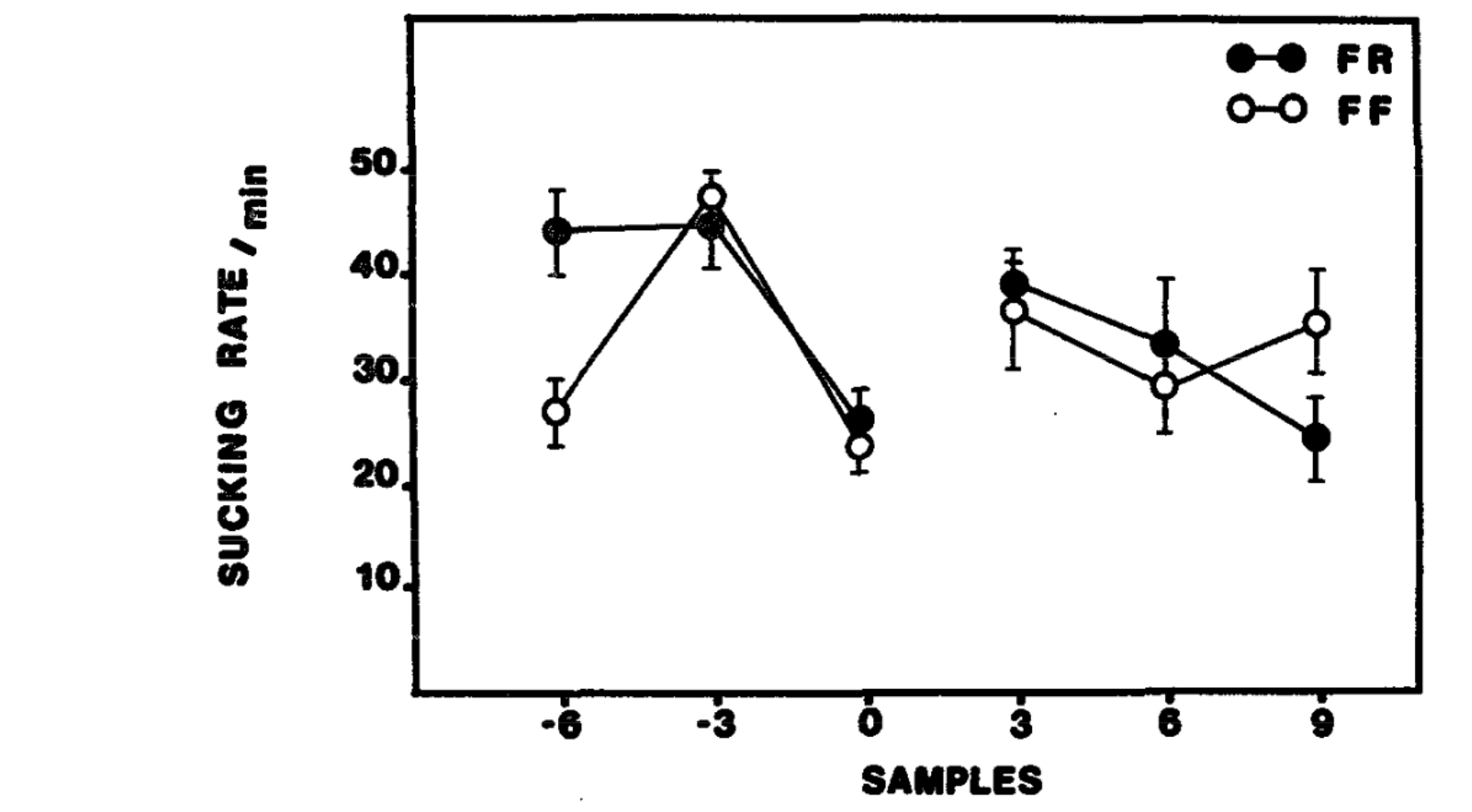


A closer look at Experiment 1 (Mehler et al. 1988)

Figure 2. *The top panel displays the sucking rates of the infants who heard Russian during the first phase of Experiment 1. Group RF heard French during the second phase while group RR heard Russian. The comparable data for subjects who heard French during the first phase are shown in the bottom panel. Group FR heard Russian during the second phase while group FF continued to hear French. The bars above and below each point indicate the standard error of the mean.*



A closer look at Experiment 1 (Mehler et al. 1988)



Interpretation of results:

- ✧ Infants prefer listening to French (Fig. 1)
- ✧ RF group has post-change increase in sucking relative to control RR: Novelty + preference for French.
- ✧ FR group doesn't have post-change increase in sucking relative to control FF: Novelty of R (increase) counteracted by dispreference for F (decrease).

Infants distinguish between their native language and a foreign language as early as 4 days old.

Experiments 2, 3 & 5

The preference that 4 day olds had for French over Russian is not due to French being nicer to listen to than Russian.

4 day old and 2 month old infants do not distinguish between two foreign languages, in general:

- 4 day old French infants don't distinguish between English and Italian.

  With Experiment 4, this isn't because English and Italian are too close to one another to be distinguished.
- 4 day old non-French infants don't distinguish between French and Russian, even though they were born in France.
- 2 month old non-French infants don't distinguish between French and Russian.

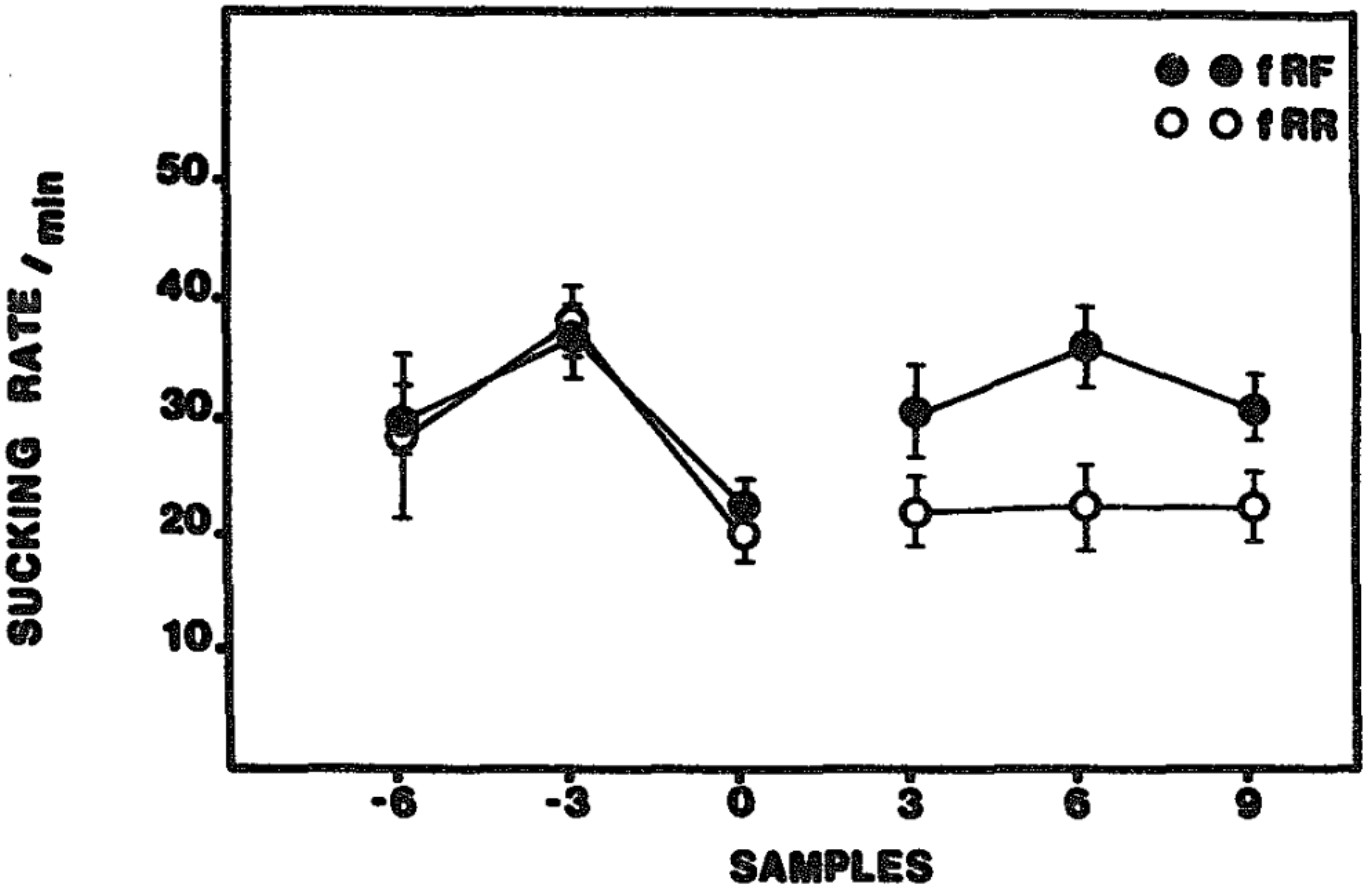
Experiments 6, 7 & 8

What in the speech signal allows infants to distinguish their first language from foreign ones?

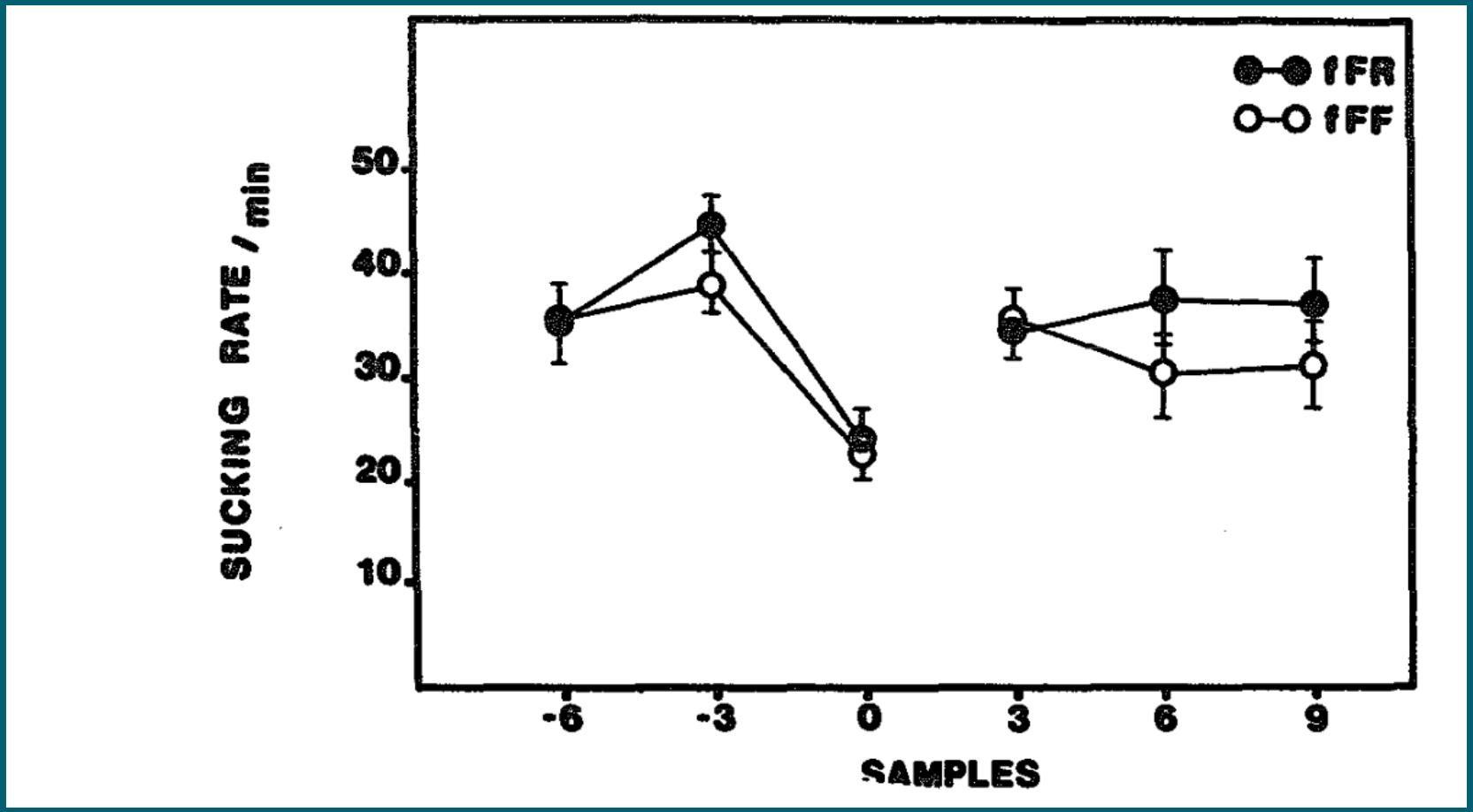
- It's not spectral information.
- Prosody is at least one factor that infants can rely on.

A closer look at Experiment 7 (Mehler et al. 1988)

Figure 8. *The top panel displays the results of the infants who heard the low-pass filtered versions of the Russian samples during the first phase of Experiment 7. Group fRF heard French during the second phase while Group fRR continued to hear Russian. The comparable data for those infants who heard the filtered French samples during the first phase are shown in the bottom panel. Group fFR heard French during the second phase while Group fFF continued to hear French.*



A closer look at Experiment 7 (Mehler et al. 1988)



Main takeaway

"[T]he speed and facility with which infants detect characteristic features of the native language suggest an innately guided learning process."

Low pass filter demo

The speech signal is made up of sounds of different frequencies.

A “low pass filter” removes higher frequency components.

Because prosodic information is mostly encoded in the lower frequencies, that remains after the filter. Most segmental information is lost.

A low pass filter is intended to mimic what can be perceived from the speech signal in utero.

Let’s see how good we are at identifying different languages under the same conditions as the ones created by the authors…

Hint: The languages I’ve brought in today are